

Deep Generative Models (ESE 6450)

Fall Semester 2026

René Vidal

Director of the Center for Innovation in Data Engineering and Science (IDEAS)

Rachleff University Professor, University of Pennsylvania

Amazon Scholar & Chief Scientist at NORCE



Course Information: Administrative



Instructor: René Vidal
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TA: Yuyan Ge
yyge@upenn.edu



TA: Buyun Liang
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TA: Nghia Nguyen
nghianhh@upenn.edu

Lectures:

Tuesday & Thursday 1:45pm-3:14pm
at AGH 203 (8/25 to 12/3)

Office Hours:

Thursday 5pm-5:45pm at AGH 609
starting 9/3



TA: Leandro Palma
palmalea@upenn.edu



TA: Uday Kiran Reddy Tadipatri
ukreddy@upenn.edu

Office Hours

- Professor René Vidal:
 - Thursday 5pm-5:45pm at AGH 609 starting 09/04
- TAs will rotate office hours every two weeks
 - Yuyan & Leandro:
 - Friday 9am – 10am. 08/28, 09/11, 09/25, 10/09, 10/23, 11/06, 11/20, 12/04
 - Buyun & Nghia:
 - Tuesday 3:30pm – 4:30pm. 09/01, 09/15, 09/29, 10/13, 10/27, 11/10, 11/24, 12/08
 - Location: AGH 615
- TAs will hold extra office hours on the week that assignments are due.
- For extra OHs with René, feel free to email Sonia Castro Rodríguez soniacr@engineering.upenn.edu

Course Website

- Please check course website for public materials:
 - <https://vidal-lab.github.io/dgm/>



University of Pennsylvania
Deep Generative Models
Fall 2026

[HOME](#) [SCHEDULE](#) [LECTURES](#) [ASSIGNMENTS](#) [PROJECT](#) [MATERIALS](#)

Deep Generative Models

Fall 2026

Updates

Course Description

Generative models have found widespread applications in science and engineering. Recent progress in deep learning has enabled the application of generative models to complex high-dimensional data such as images, videos, text and speech. This course will cover state-of-the-art deep generative models, including variational autoencoders (VAEs), auto-regressive models, diffusion models, and generative adversarial networks (GANs). The course will also illustrate various applications of deep generative models to image and video generation, text and speech generation, image captioning, text-to-image generation, and inverse problems.



Prof. René Vidal



Uday Kiran Reddy Tadipatri



Yuyan Ge



Buyun Liang



Nghia Nguyen



Leandro Palma



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[HOME](#) [SCHEDULE](#) [LECTURES](#) [ASSIGNMENTS](#) [PROJECT](#) [MATERIALS](#)

Course Schedule

Class Time: Tuesdays and Thursdays, 3:30 PM - 4:59 PM
Location: AGH 203
Semester Dates: 8/25/2026 - 12/3/2026
Office Hours: Professor René Vidal: Thursday 5pm-5:45pm, AGH 609, starting 09/04
TAs will rotate office hours every two weeks @ AGH 610:

- Yuyan & Leandro: Friday 9am-10am, 08/28, 09/11, 09/25, 10/09, 10/23, 11/06, 11/20, 12/04
- Buyun & Nghia: Tuesday 3:30pm-4:30pm, 09/01, 09/15, 09/29, 10/13, 10/27, 11/10, 11/24, 12/08

TAs will hold extra office hours on the week that assignments are due.
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Event	Date	Title	Topics
Lecture	Aug 26, 2025 Tuesday	Introduction	Introduction
Lecture	Aug 28, 2025 Thursday	Background	Probability, Statistics, MLE, SGD, Learning
Lecture	Sep 02, 2025 Tuesday	Latent Variable Models	Gaussian Models, PPCA
Lecture	Sep 04, 2025 Thursday	Latent Variable Models	PPCA
Lecture	Sep 09, 2025 Tuesday	Latent Variable Models	Variational Inference
Lecture	Sep 11, 2025 Thursday	Latent Variable Models	EM, EM for GMMs, VAE (definition)
Lecture	Sep 16, 2025 Tuesday	Latent Variable Models	VAE (parameterization trick) - application + hands on session
Lecture	Sep 18, 2025 Thursday	Shallow Auto-regressive Models	Markov Models
Lecture	Sep 23, 2025 Tuesday	Shallow Auto-regressive Models	HMMs (Inference and Decoding)
Lecture	Sep 25, 2025 Thursday	Recitation	Hands-on VAE
Lecture	Sep 30, 2025 Tuesday	Shallow Auto-regressive Models	HMMs (Learning, Examples)

Course Materials

Prerequisites

Students should be comfortable with **multivariate calculus**, **linear algebra**, **probability**, **statistics**, **information theory**, and **Python programming** to register for the course.

Basic knowledge of **machine learning** (e.g., ESE 4200, CIS 5190, or CIS 5200) would be very useful.

Reference Books

Generalized Principal Component Analysis

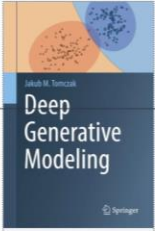
René Vidal, Yi Ma, Shankar Sastry

1st Edition ISBN: 978-0-387-87810-2

Deep Generative Modeling

Jakub M. Tomczak

1st Edition ISBN: 978-3-030-93157-5



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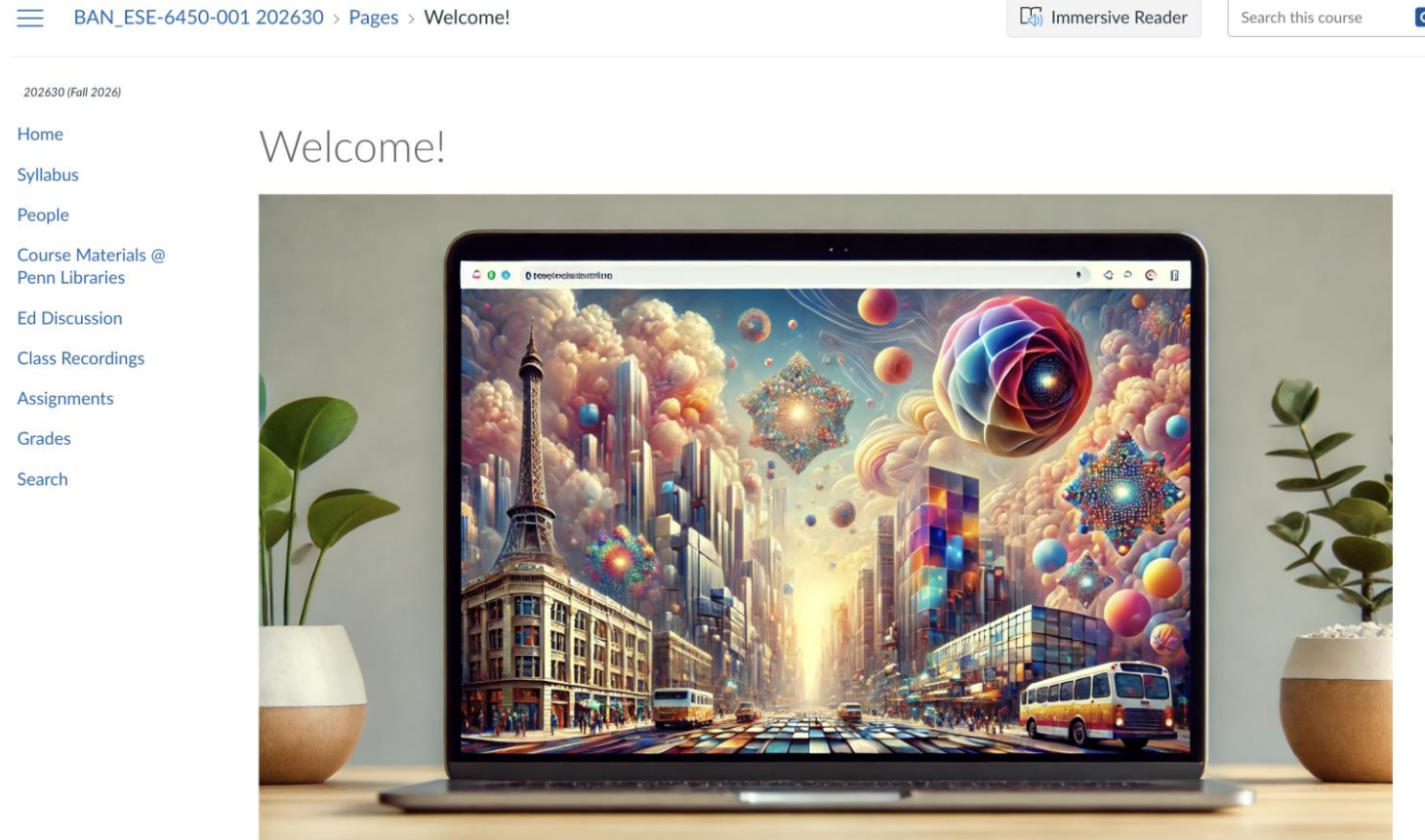
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Course Canvas

- Please check course Canvas for all materials:
 - <https://canvas.upenn.edu/courses/1937189>



DALL-E 2024-08-26 15.16.34 - A vibrant and artistic representation of deep generative models for a website cover photo. The image should include a visually captivating display of .webp

ESE 6450-001: Deep Generative Models

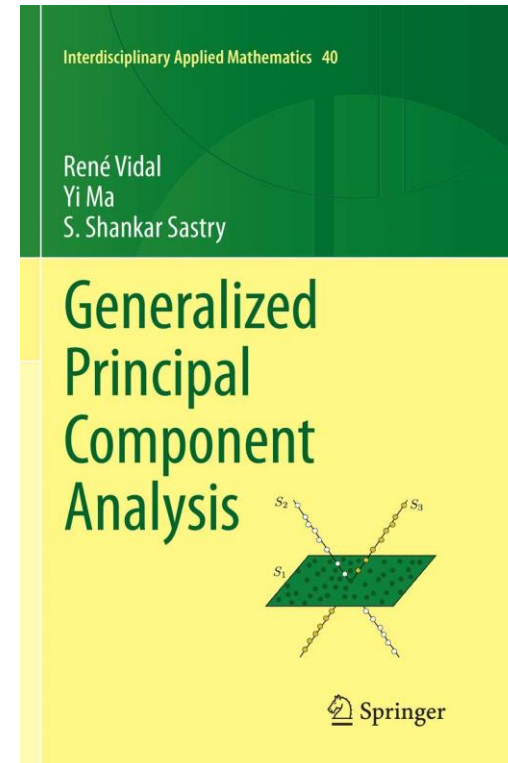
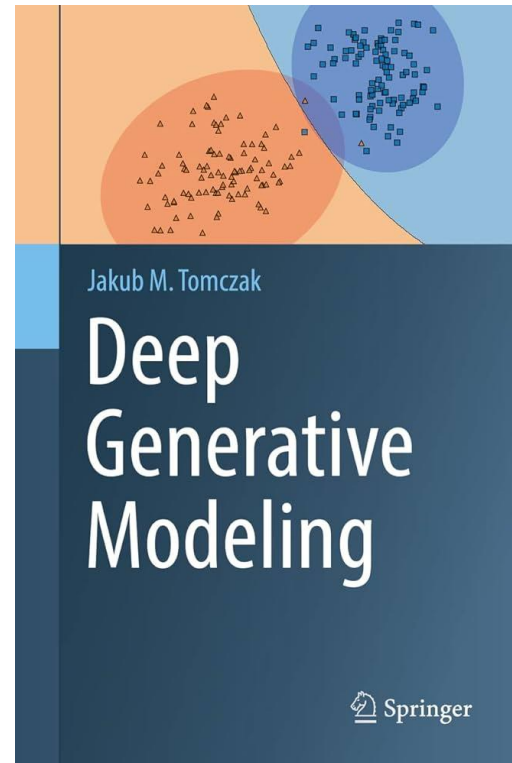
Please see Course Website for further information. <https://vidal-lab.github.io/dgm/>

Course Information: Background

- Graduate-level course
- **Prerequisites:** Students should be comfortable with
 - **Linear algebra:** matrices, (pseudo)-inverses, positive definiteness, eigenvalue decomposition, singular value decomposition, matrix-valued derivatives
 - **Multivariate calculus, optimization:** gradient, Hessian, first and second order conditions for minima/maxima, gradient descent, alternating minimization, Lagrange Multipliers
 - **Probability, statistics, information theory:** random variables, expectation, variance, covariance, maximum likelihood, expectation maximization, entropy, mutual information
 - **Programming:** Python
- Prior exposure to **machine learning** (e.g., ESE 4200, CIS 5190 or CIS 5200) is a plus.

Course Information: Books and Grading

- Text(s)/Required Materials:
 - Deep Generative Modeling, Jakub M. Tomczak, Springer Verlag, 2022 (<https://link-springer-com.proxy.library.upenn.edu/book/10.1007/978-3-030-93158-2>)
 - Generalized Principal Component Analysis, René Vidal, Yi Ma, Shankar Sastry, Springer Verlag, 2016 (<https://link-springer-com.proxy.library.upenn.edu/book/10.1007/978-0-387-87811-9>)



Course Syllabus

- Introduction (Week 1)
 - History of AI, History of Generative Models (GMs), Advent of Deep GMs, Applications
- Background (Week 1, 2)
 - Basics of Linear Algebra, Probability, Statistics, Information Theory, and Optimization
 - Discriminative vs Generative Models
 - Taxonomy of Generative Models
 - Maximum Likelihood Estimation
- Latent Variable Models (Week 2, 3, 4)
 - Probabilistic Principal Component Analysis (PPCA)
 - Application of PPCA to Generating Images of Faces with Variable Lighting
 - Variational Inference and Expectation Maximization
 - Variational Auto-Encoders (VAEs)
 - Application of VAEs to Generating Images of Handwritten Digits

Course Syllabus

- Shallow Auto-regressive Models (Week 4, 5, 6, 7, 8)
 - Markov Models & Hidden Markov Models (HMMs)
 - Application of HMMs to Generating Text
 - Linear Dynamical Systems (LDSs)
 - Application of LDSs to Generating Videos of Dynamic Textures
- Deep Auto-regressive Models (Week 8, 9, 10, 11, 12, 13)
 - Recursive Neural Networks (RNNs)
 - Application to Speech Synthesis (WaveNet) and Image Captioning (RNN + VAE)
 - Diffusion Models
 - Denoising Diffusion Probabilistic Models (DDPM)
 - Diffusion Models with Latent Variables
 - Implementation details (UNet, StyleGAN)
 - Applications to Image Inpainting, and Text-to-Image Generation (CLIP Guided Diffusion)
 - Transformers (Transformer, Vision Transformer, Multimodal Transformers)
 - Application to Text Generation, and Text-to-Image Generation (DALLE: auto-regressive model + VAE)

Course Syllabus

- Multimodal Models (Week 13, 14, 15)
 - Vision-Language Models
 - ControlNet
 - Applications to Image Editing

Grading

- Assignment Quizzes: 60%
 - Three mini-project assignments covering different paradigms of generative models.
 - Due on the last week of the month, likely on September 29, October 27, and November 24.
 - Each assignment must be completed and submitted but will not be graded directly.
 - The grade for each assignment unit will be determined by an in-class quiz on the assignment due date, testing the student's understanding of the submitted assignment.
- Final Project Exam: 40%
 - Team project on text-to-image generation and editing.
 - Due on the University-assigned final-exam date: to be announced on October 2, 2026.
 - A project report must be completed and submitted but will not be graded directly.
 - The project grade will be determined by an individual final exam testing the student's understanding of multimodal generative models and of their team project, including its methods, implementation, experiments, results, and limitations.

Use of AI Tools

- Generative AI tools may be used while completing the assignments and final project to help understand concepts, derive results, write and debug code, analyze experiments, improve explanations, and explore alternative approaches.
- However, the purpose of the assignments and project is to support learning. You are responsible for understanding the materials, derivations, code, experiments, and conclusions you submit, even if AI tools were used to produce them.
- Submitted assignments and the project report will not be graded directly. Your understanding will be evaluated individually through quizzes and exams, where you will be asked to explain, derive, modify or reason about your submitted work.
- AI tools may not be used during in-class quizzes or the Final Project Exam.
- You should treat AI-generated output critically: it can have incorrect derivations, faulty code, misleading explanations, conclusions inconsistent with experiments.
- In short, you are welcome to use AI as a tool for learning, but you should be prepared to independently explain and reason about everything you submit.

Student Code of Ethics

- Read University Policy:
 - <https://grad.seas.upenn.edu/student-handbook/student-code-of-conduct/>
- The Penn Engineering student is expected to represent themselves honestly in all oral or written statements [and] to represent their academic product honestly and fairly. The student will not knowingly use any dishonest method to gain an unfair advantage over other students in academic pursuits, especially through, but not limited to:
 - Giving or receiving any unauthorized aid on an assignment or exam, including working in groups on any assignment that has been designated as individual by the professor;
 - Misrepresenting the originality of one's work (plagiarism), particularly through failing to footnote the contributions of another, except as permitted by the instructor;
 - Continuing to write after time has been called on an exam; or
 - Submitting substantially the same work for credit in more than one class, except with prior approval of the instructor.

Registration & Auditing

- Number of registered students: ~85
- Number of students in the waiting list: ~61
- Capacity of AGH 203: ~85
- I'm really sorry since a larger room is not available, we will not be able to accept auditing students. All students must be registered for credit.